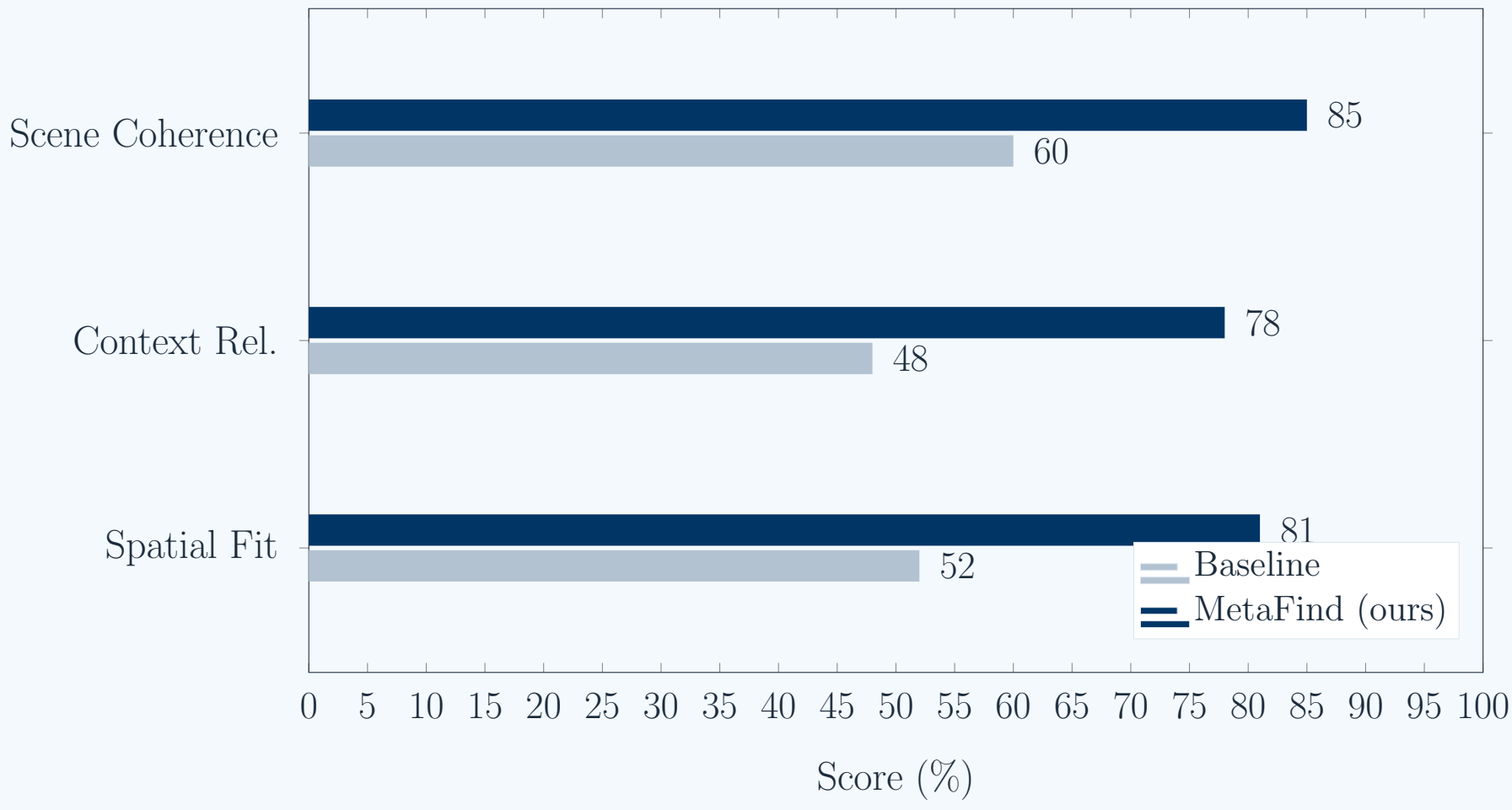


- User studies confirm that contextually coherent assets significantly improve perceived immersion and realism.
- The pipeline generalises across domain types: interiors, outdoor environments, and abstract virtual worlds.



## Conclusion

- Scene-awareness** is critical for high-quality automated 3D scene generation in the Metaverse.
- MetaFind outperforms traditional retrieval approaches by **+25%** on coherence and spatial fit.
- The modular pipeline is scalable and can integrate with existing Metaverse rendering engines.

## Future Work

- Real-time rendering integration
- Deep learning scene graph models
- Larger 3D asset datasets
- Multi-modal prompting (sketch + text)
- User-guided scene refinement
- Cross-platform VR support

## References

[1] Ma et al. — *SceneComposer: 3D scene synthesis from language*, 2023.

[2] Chen et al. — *Embedding-based 3D model retrieval*, CVPR 2022.

[3] Park & Kim — *Foundations of Metaverse environments*, 2022.

[4] Liu et al. — *Scene graph approaches for layout generation*, 2023.